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part2pop: a modular framework for standardizing and analyzing aerosol particle populations

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Summary

Aerosol particles are chemically diverse, multicomponent systems that vary widely in size, composition, and mixing state. These particle-scale differences strongly influence cloud condensation nuclei activity, ice nucleation, and optical properties (Alpert & Knopf, 2016; Cubison et al., 2008; Fierce et al., 2013, 2016, 2020; Lata et al., 2021; Petters & Kreidenweis, 2007; Riemer et al., 2019).

Across models, observations, and parameterized descriptions, aerosol populations are represented in incompatible ways. Different representations preserve different aspects of aerosol size, composition, and mixing state, which limits direct comparison across representations (Fierce et al., 2024; Liu et al., 2016; Riemer et al., 2009; Stevens & Dastoor, 2019; Tilmes et al., 2023). As a result, similar analyses are frequently reimplemented for each representation, making comparison and reuse time-consuming.

`part2pop` is a lightweight Python library that standardizes heterogeneous aerosol populations into a common data model. The package supports population construction from model output, parameterizations, and observation-constrained reconstructions, and it provides reusable particle- and population-level diagnostics for aerosol composition, hygroscopicity, cloud activation, optical properties, and freezing. Its registry-based builder architecture makes the package extensible, allowing new species definitions, population types, particle morphologies, and diagnostic variables to be added without modifying the core representation.

Statement of need

Aerosol populations from particle-resolved models, reduced representations such as modal and sectional models, parameterized distributions, and observation-based reconstructions represent different information about particle size, composition, abundance, and mixing state (Bauer et al., 2008; DeCarlo et al., 2006; Jayne et al., 2000; Liu et al., 2016; Mann et al., 2012; Riemer et al., 2009; Stevens & Dastoor, 2019; Tilmes et al., 2023; Wang, Pikridas, Spielman, et al., 2017; Zaveri et al., 2008; Zelenyuk et al., 2015). Consequently, these populations cannot be directly compared without translation, and analyses such as composition metrics, CCN activity, optical properties, and freezing diagnostics are often reimplemented for each representation.

`part2pop` addresses this gap by providing a shared aerosol-population data model and analysis interface that makes such analyses reusable across heterogeneous sources. The package does not introduce a new aerosol simulation model. Instead, it standardizes populations derived from simulations, observations, and parameterizations so that they can be analyzed, compared, and reused consistently. Intended users include aerosol model developers, instrument developers and observationalists, process modelers requiring aerosol inputs, instructors and students, and researchers performing intercomparisons or sensitivity analyses. The need addressed is not a lack of aerosol microphysics models, but a lack of a common internal representation of

aerosol populations that makes existing tools reusable across models, parameterizations, and observations.

State of the field

Existing software in aerosol science typically focuses on either source-specific aerosol populations or targeted downstream calculations, rather than interoperability across representations. Aerosol simulation frameworks generate populations in source-specific representations, reflecting trade-offs between physical detail and computational efficiency (Liu et al., 2016; Mann et al., 2012; Riemer et al., 2009; Tilmes et al., 2023). Observational workflows constrain different subsets of aerosol population properties through instrument-specific measurements (Chen et al., 2013; Cheng et al., 2022; DeCarlo et al., 2006; Krueger et al., 2003; Wang, Pikridas, Spielman, et al., 2017; Zelenyuk et al., 2015). Other tools then target specific diagnostics such as optical properties, cloud activation, or ice nucleation (Alpert & Knopf, 2016; Petters & Kreidenweis, 2007; Sumlin et al., 2018). However, these components do not share a common internal representation of aerosol populations, which limits reuse of downstream analyses across sources. As a result, populations derived from different sources cannot be readily combined, compared, or analyzed using shared diagnostics.

`part2pop` complements rather than duplicates these tools. Its distinct contribution is a shared representation layer that standardizes heterogeneous aerosol populations and enables the same downstream diagnostics to be applied across them.

Software design

`part2pop` adopts a config-driven architecture centered on a small set of core data structures and registry-discovered builders (Figure 1). The primary structures are `AerosolSpecies`, `Particle`, and `ParticlePopulation`. `AerosolSpecies` defines intrinsic species properties such as density, molar mass, hygroscopicity, and surface-tension defaults, with optional overrides applied at build time. `ParticlePopulation` stores a species list, species-resolved particle masses, number concentrations, and particle identifiers. Individual particles are accessed and modified through the population interface. Together, these structures define a unified internal representation of aerosol composition, size, and abundance across heterogeneous sources.

The core representation is mass-based. This is a deliberate design trade-off: it is general enough to accommodate heterogeneous population sources while remaining simple enough to support consistent derived-property calculations. Population construction is handled by `PopulationBuilder` using configuration dictionaries (`population_cfg`), and diagnostic construction is handled by `VariableBuilder` using `var_cfg`. Both rely on registry-backed factory discovery, so new population builders and diagnostic modules can be added without modifying the core classes.

Current population types include model-derived populations from the Particle Monte Carlo model (`partmc`) and the four-mode Modal Aerosol Module (`mam4`) (Liu et al., 2016; Riemer et al., 2009), parameterized populations such as `monodisperse`, `binned_lognormals`, and `sampled_lognormals`, and observation-based populations such as `hiscale_observations` and `edx_observations`. The `hiscale_observations` builder reconstructs populations from complementary measurements collected during the HI-SCALE field campaign (Fast et al., 2019), including Fast Integrated Mobility Spectrometer (FIMS) size distributions, Aircraft-Integrated Meteorological Measurement System (AIMMS) aircraft-state data, miniSPLAT single-particle mass-spectrometer composition classes, and Aerosol Mass Spectrometer (AMS) bulk-composition constraints (Beswick et al., 2008; DeCarlo et al., 2006; Jayne et al., 2000; Wang, Pikridas, Spielman, et al., 2017; Wang, Pikridas, Pinterich, et al., 2017; Zelenyuk et al., 2015). The `edx_observations` builder reconstructs populations from particle size, class, and elemental-composition measurements collected on substrates using the Size and

Time-Resolved Aerosol Collector (STAC) and analyzed using scanning electron microscopy with energy-dispersive X-ray spectroscopy (SEM-EDX) or high-resolution focused ion beam/scanning electron microscopy (FIB/SEM) chemical imaging (Chen et al., 2013; Cheng et al., 2022; Krueger et al., 2003). STAC has been demonstrated for vertical aerosol sampling on the Atmospheric Radiation Measurement (ARM) user facility Tethered Balloon System (TBS) (Cheng et al., 2022).

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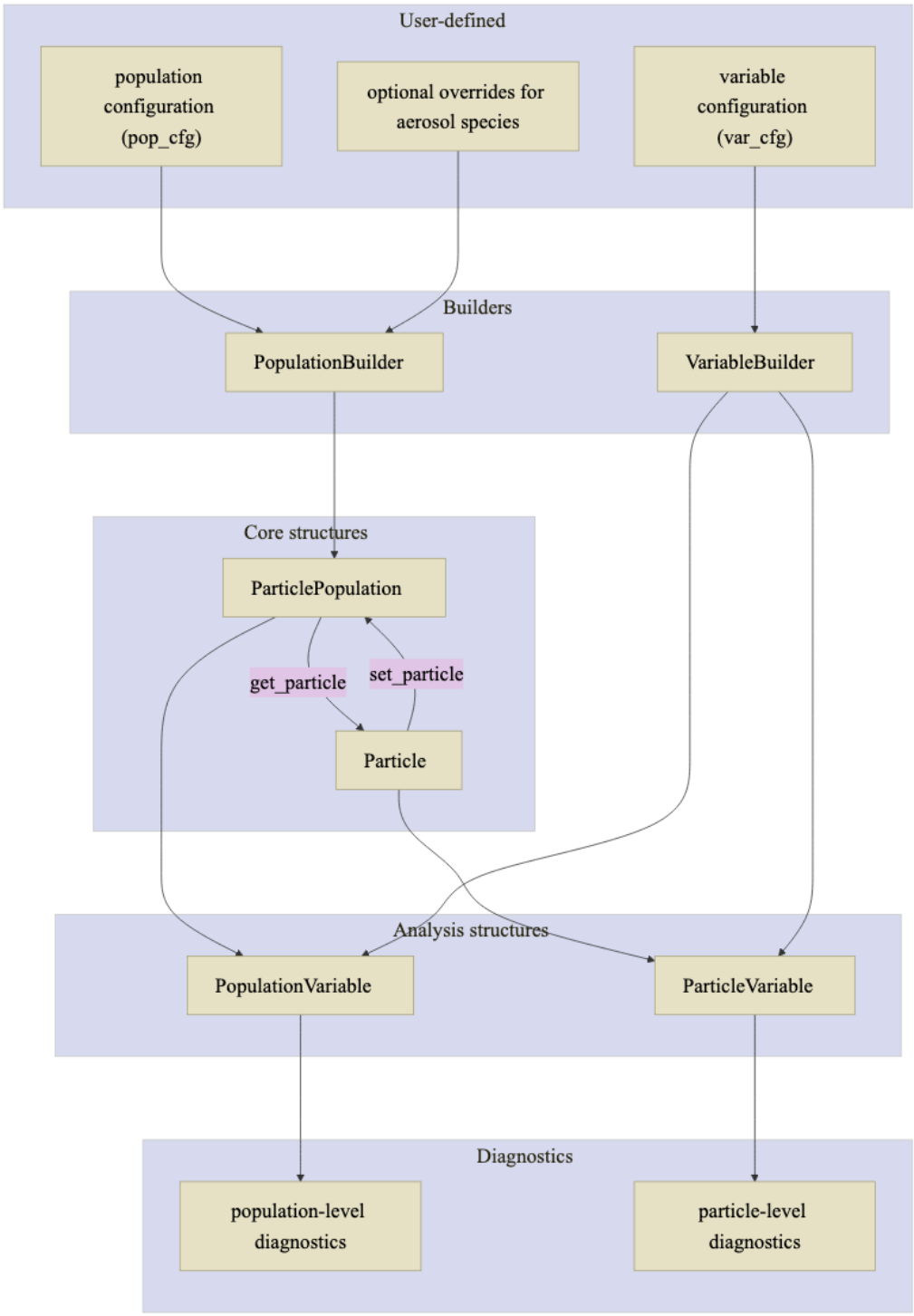

Figure 1: Overview of the config-driven architecture in `part2pop`. User-defined configuration files are processed by builder modules to construct standardized particle populations and derived analysis variables. Particle- and population-level diagnostics are then computed from these shared representations.

Process-specific extensions such as `OpticalPopulation` and `FreezingPopulation` wrap the same base population representation rather than reimplementing population logic. This allows optical and freezing calculations to reuse the same underlying particle and population definitions

while interfacing with established frameworks such as κ-Köhler theory and Mie scattering implementations (Alpert & Knopf, 2016; Petters & Kreidenweis, 2007; Sumlin et al., 2018). A visualization module and optional graphical viewer support exploratory analysis.

The architecture is also intentionally designed to be modifiable by both human developers and automated code-generation systems. Config-driven workflows, registry-based module discovery, minimal coupling between core data structures and analysis modules, and stable builder interfaces localize code changes. As a result, new population builders or diagnostics can be added without invasive edits to the core package, lowering the barrier to extension for both human developers and AI-assisted code generation.

Research impact statement

`part2pop` targets an active methodological bottleneck in aerosol science: representation choice itself can introduce structural differences in aerosol diagnostics and can complicate comparison across models and datasets (Fierce et al., 2017, 2024; Fierce & McGraw, 2017). By making particle-resolved, modal, sectional, parameterized, and observation-based populations analyzable through a shared interface, the package provides reusable infrastructure for cross-representation comparison, diagnostic reuse, and initialization of downstream models.

The current release includes reproducible workflows for optical, CCN-relevant, freezing-related diagnostics for aerosol particle populations from a variety of sources. This positions `part2pop` as reusable research infrastructure rather than study-specific analysis code and supports its use in intercomparison and observation-constrained aerosol analysis.

AI usage disclosure

The core architecture, data model, and software design were developed by the authors without the use of generative AI tools. After the core structure was established, OpenAI's ChatGPT (versions 5.1–5.4) was used for code development and refactoring, development of the graphical user interface, and drafting and editing of documentation and manuscript text.

All AI-generated code and text were reviewed, modified, and verified by the authors, validated through testing and example workflows, and checked for scientific correctness. The software architecture was intentionally designed to support modification by automated code-generation tools, including large language models, through its modular and configuration-driven structure.

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